

Phase 2 of Traffic light controller project

NOTE: This is just a planning flow that might be changed during development, Phase 2 is currently under development.

Existing System

file name and their work

1. timer.v - Generate time, 1 clk cycle is equal to 1 second.
2. state_register.v - Store the current state of the system.
3. Next_state.v - Based on current state and inputs, decide the next state and set the priority.
4. output_logic.v - Based on the current state, decide which output pin should go high and low.
5. traffic_light_controller.v - use all the modules to build the system (top level module)

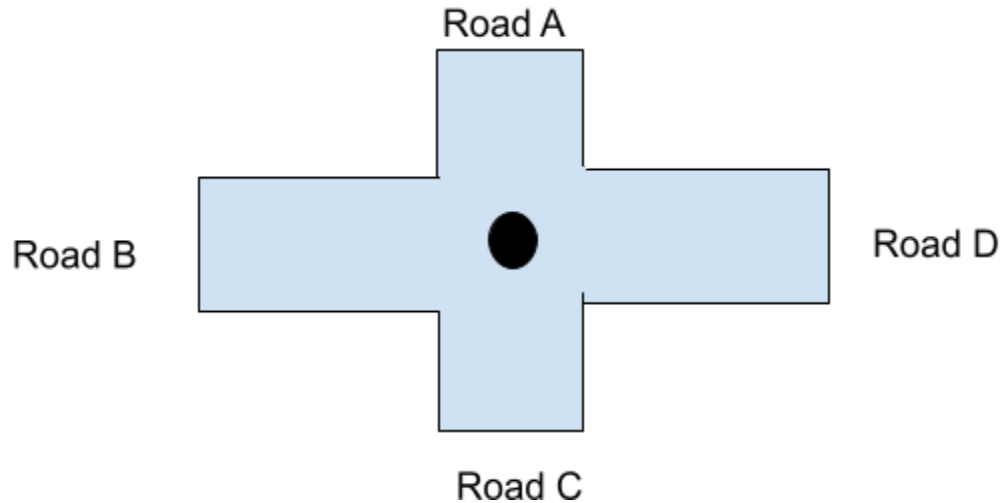
State table

State	Name
S0	NS_Green
S1	NS_Yellow
S2	All_red_NS
S3	walk_NS
S4	Emergency_NS
S5	EW_Green
S6	EW_yellow
S7	All_red_EW
S8	Walk_EW
S9	Emergency_EW

Phase 2 -

Next System Requirements

1. Four-Way traffic management
2. Expanded FSM architecture
3. Additional pedestrian crossings
4. Emergency handling for four direction
5. More advanced traffic sequencing.



Input and output of the system

- **Inputs**
 - Clk - 1 bit
 - Reset - 1 bit
 - emergency_A - 1 bit
 - ped_req_A - 1 bit
 - emergency_B - 1 bit
 - ped_req_B - 1 bit
 - emergency_C - 1 bit
 - ped_req_C - 1 bit
 - emergency_D - 1 bit
 - ped_req_D - 1 bit
- **Outputs**
 - Red_A - 1 bit
 - Green_A - 1 bit
 - Yellow_A - 1 bit
 - wlak_A - 1 bit
 - dont_walk_A - 1 bit
 - Red_B - 1 bit
 - Green_B - 1 bit
 - Yellow_B - 1 bit
 - wlak_B - 1 bit
 - dont_walk_B - 1 bit

- Red_C - **1 bit**
- Green_C - **1 bit**
- Yellow_C - **1 bit**
- wlak_C - **1 bit**
- dont_walk_C - **1 bit**
- Red_D - **1 bit**
- Green_D - **1 bit**
- Yellow_D - **1 bit**
- wlak_D - **1 bit**
- dont_walk_D - **1 bit**

(Total 10 input signal and 20 output signal)

Internal signals

1. Current state - **5 bit**
2. Next state - **5 bit**
3. Timer done - **1 bit**
4. Target count - **5 bit**
5. Timer clear - **1 bit**

State table

State	Name
S0	A_Green
S1	A_Yellow
S2	All_red_A
S3	walk_A
S4	Emergency_A
S5	B_Green
S6	B_yellow
S7	All_red_B
S8	Walk_B
S9	Emergency_B
S10	C_Green

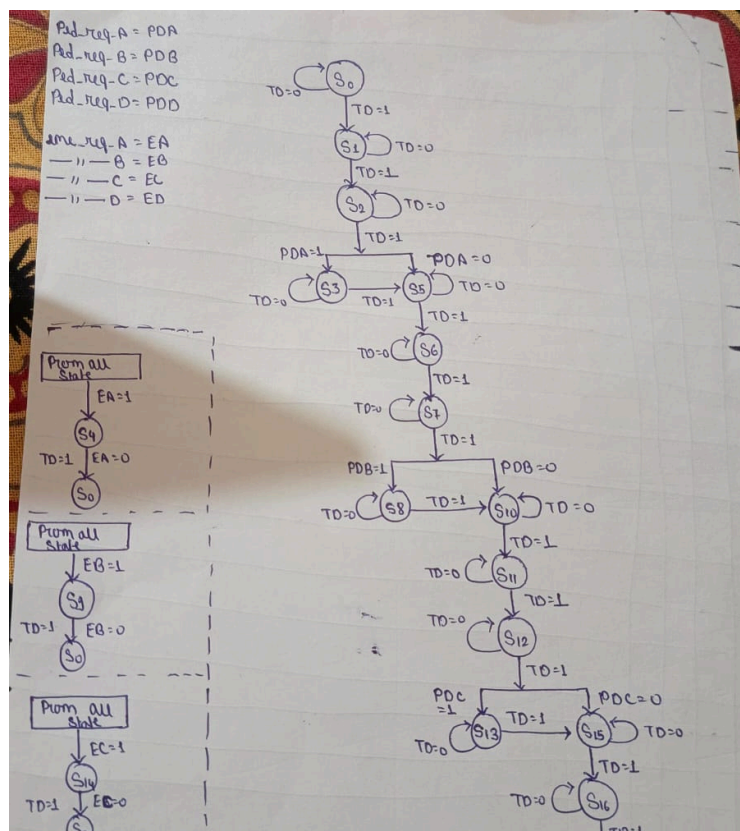
State	Name
S11	C_Yellow
S12	All_red_C
S13	walk_C
S14	Emergency_C
S15	D_Green
S16	D_Yellow
S17	All_red_D
S18	walk_D
S19	Emergency_D

Total 20 states form S0 to S19

Important points

1. Normal system flow $S0 > S1 > S2 > S5 > S6 > S7 > S10 > S11 > S12 > S15 > S16 > S17 > S0$. (With no ped req and emergency request).
2. Ped request should check on state S2, S7, S12, S17.
3. Emergency requests should check on every state with priority $A > B > C > D$.

State diagram



Output Truth table

State	A_red	A_green	A_yellow	A_walk	A_dont_walk	B_red	B_green	B_yellow	B_walk	B_dont_walk	C_red	C_green	C_yellow	C_walk	C_dont_walk	D_red	D_green	D_yellow	D_walk	D_dont_walk
S0	0	1	0	0	1	1	0	0	0	1	1	0	0	0	1	1	0	0	0	1
S1	0	0	1	0	1	1	0	0	0	1	1	0	0	0	1	1	0	0	0	1
S2	1	0	0	0	1	1	0	0	0	1	1	0	0	0	1	1	0	0	0	1
S3	1	0	0	1	0	1	0	0	0	1	1	0	0	0	1	1	0	0	0	1
S4	0	1	0	0	1	1	0	0	0	1	1	0	0	0	1	1	0	0	0	1
S5	1	0	0	0	1	0	1	0	0	1	1	0	0	0	1	1	0	0	0	1
S6	1	0	0	0	1	0	0	1	0	1	1	0	0	0	1	1	0	0	0	1
S7	1	0	0	0	1	1	0	0	0	1	1	0	0	0	1	1	0	0	0	1
S8	1	0	0	0	1	1	0	0	1	0	1	0	0	0	1	1	0	0	0	1
S9	1	0	0	0	1	0	1	0	0	1	1	0	0	0	1	1	0	0	0	1
S10	1	0	0	0	1	1	0	0	0	1	0	1	0	0	1	1	0	0	0	1
S11	1	0	0	0	1	1	0	0	0	1	0	0	1	0	1	1	0	0	0	1
S12	1	0	0	0	1	1	0	0	0	1	1	0	0	0	1	1	0	0	0	1
S13	1	0	0	0	1	1	0	0	0	1	1	0	0	1	0	1	0	0	0	1
S14	1	0	0	0	1	1	0	0	0	1	0	1	0	0	1	1	0	0	0	1
S15	1	0	0	0	1	1	0	0	0	1	1	0	0	0	1	0	1	0	0	1
S16	1	0	0	0	1	1	0	0	0	1	1	0	0	0	1	0	0	1	0	1

State	A_red	A_green	A_yellow	A_walk	A_dont_walk	B_red	B_green	B_yellow	B_walk	B_dont_walk	C_red	C_green	C_yellow	C_walk	C_dont_walk	D_red	D_green	D_yellow	D_walk	D_dont_walk
S17	1	0	0	0	1	1	0	0	0	1	1	0	0	0	1	1	0	0	0	1
S18	1	0	0	0	1	1	0	0	0	1	1	0	0	0	1	1	0	0	1	0
S19	1	0	0	0	1	1	0	0	0	1	1	0	0	0	1	0	1	0	0	1

Files need to modified in phase 2

1. State_register_logic.v
2. Next_state_logic.v
3. Output_logic.v
4. traffic_light_controller.v

(also their respective TB for module testing)

State encoding for phase 2

State	Encoding
S0	00000
S1	00001
S2	00010
S3	00011
S4	00100
S5	00101
S6	00110
S7	00111
S8	01000
S9	01001
S10	01010

State	Encoding
S11	01011
S12	01100
S13	01101
S14	01110
S15	01111
S16	10000
S17	10001
S18	10010
S19	10011